



# Amazon Machine Images (AMI)

## Lecture Slides

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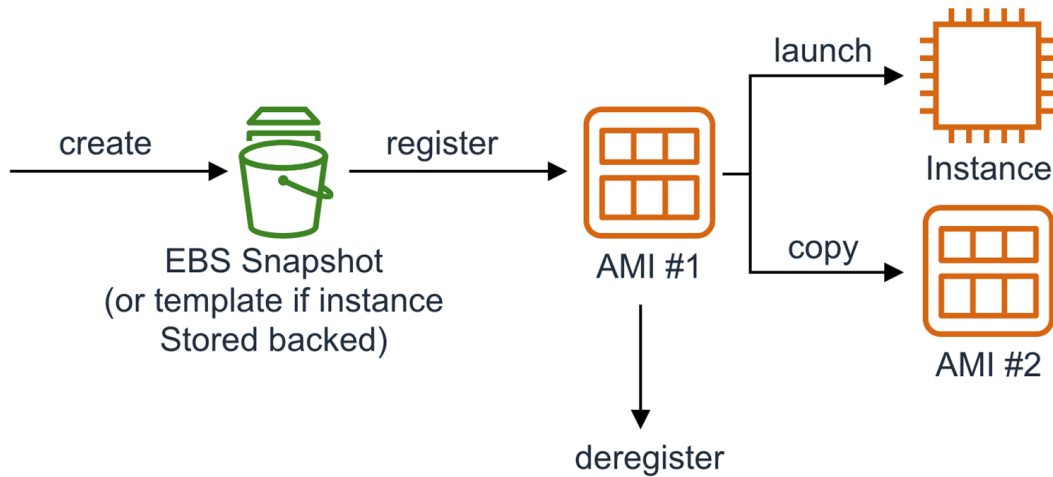
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Good luck on your exam!

# EC2 - AMI



**Amazon Machine Image (AMI)** provides the necessary information to launch a virtual server, or **EC2 instance**, in the AWS cloud. It includes the OS, application server, applications, and related configurations.



**An AMI holds the following information:**

- **Template** for the root volume (based on an **EBS snapshot** or **Instance Store**)
- **Launch Permissions:** Determines which AWS accounts can use the AMI to launch instances
- **Block Device Mapping:** Defines additional volumes to attach at instance launch



AMIs are **Region Specific!**

# AMI - Use Cases

AMIs supports **incremental changes** to OS, application code, and packages.



**web-server-000**

Ruby, Node, Postgres Client Installed



**web-server-001**

Redis for Sidekiq Installed



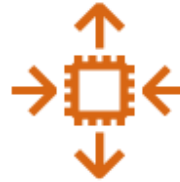
**web-server-002**

ImageMagick for Image Processing Installed



**web-server-003**

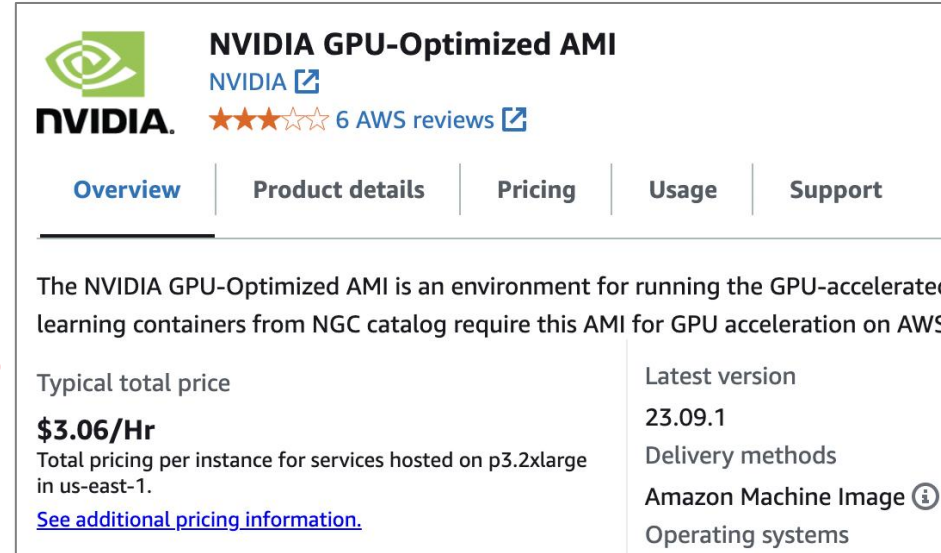
CloudWatch Agent Installed



- Automate **patching and updates** via **AWS Systems Manager Automation**
- Ideal for replicating environments across accounts and regions
- Used in **Launch Templates** or **Auto Scaling groups** to maintain consistency

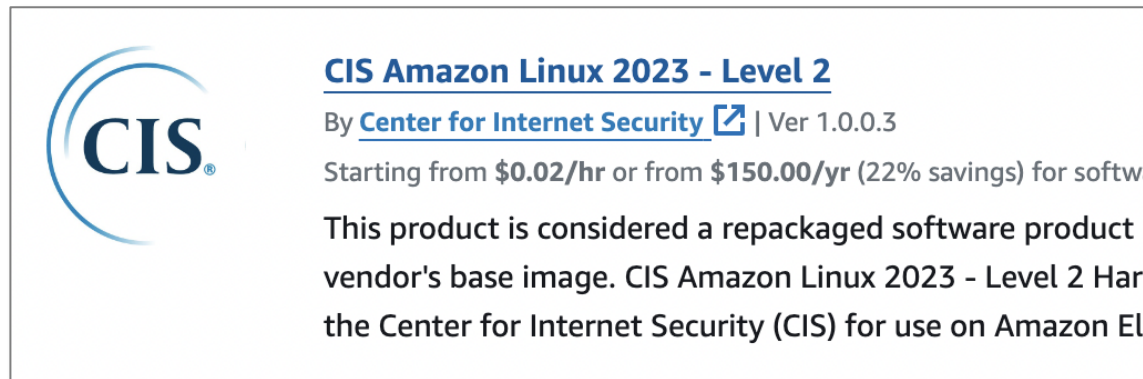
# AWS Marketplace

The AWS Marketplace lets you **purchase subscriptions** to vendor-maintained AMIs.



The screenshot shows the product page for the NVIDIA GPU-Optimized AMI. At the top, there is the NVIDIA logo and the product title "NVIDIA GPU-Optimized AMI" with a link to the NVIDIA website. Below the title, there are 6 AWS reviews with a star rating. A navigation bar includes links for Overview, Product details, Pricing, Usage, and Support. The main content area describes the AMI as an environment for running GPU-accelerated learning containers. A pricing table shows a typical total price of \$3.06/Hr for p3.2xlarge instances in us-east-1, with a link to additional pricing information. On the right, there is a sidebar with details about the latest version (23.09.1), delivery methods (Amazon Machine Image), and operating systems.

| Typical total price                                                        |                        |
|----------------------------------------------------------------------------|------------------------|
| <b>\$3.06/Hr</b>                                                           | Latest version         |
| Total pricing per instance for services hosted on p3.2xlarge in us-east-1. | 23.09.1                |
| <a href="#">See additional pricing information.</a>                        | Delivery methods       |
|                                                                            | Amazon Machine Image ⓘ |
|                                                                            | Operating systems      |



The screenshot shows the product page for CIS Amazon Linux 2023 - Level 2. It features the CIS logo and the product title. Below the title, it says "By Center for Internet Security" with a link to the CIS website and "Ver 1.0.0.3". The pricing is listed as starting from \$0.02/hr or from \$150.00/yr (22% savings) for software. The description states that this product is considered a repackaged software product based on the vendor's base image. The bottom of the screenshot is partially cut off.

**CIS Amazon Linux 2023 - Level 2**  
By [Center for Internet Security](#) | Ver 1.0.0.3  
Starting from **\$0.02/hr** or from **\$150.00/yr** (22% savings) for software  
This product is considered a repackaged software product based on the vendor's base image. CIS Amazon Linux 2023 - Level 2 Hardened is provided by the Center for Internet Security (CIS) for use on Amazon ELB.

**Security-hardened AMIs** are very popular.  
eg. Center of Internet Security (CIS)

# Choosing an AMI

It's important to note that the AMI ID will be different for each region.  
So, you need to always confirm you are using the correct regional AMI ID.

US East 1 AMI ID

Some AMI's will have multiple **architecture options**  
with an alternate AMI ID



Amazon Linux 2023 AMI

ami-0440d3b780d96b29d (64-bit (x86), uefi-preferred) / ami-0f93c02efd1974b8b (64-bit (Arm), uefi)

Amazon Linux 2023 is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Platform: amazon      Root device type: ebs      Virtualization: hvm      ENA enabled: Yes

**Select**

Amazon Linux

Free tier eligible

Verified provider

64-bit (x86), uefi-preferred

64-bit (Arm), uefi

CA-Central-1 AMI ID



Amazon Linux 2023 AMI

ami-0156b61643fdfee5c (64-bit (x86), uefi-preferred) / ami-0d498e5e3490ebfaf (64-bit (Arm), uefi)

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Free tier eligible

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64-bit (x86), uefi-preferred

64-bit (Arm), uefi

# Choosing an AMI

AMIs can be selected based on:

- **Region**
- **Operating System**
- **Architecture** (64-bit x86 or Arm)
- **Launch Permissions:** Public, Private, or Shared
- **Root Volume Type:**
  - Instance Store (Ephemeral Storage)
  - EBS Backed Volumes

## ▼ Linux/Unix

- ☐ All Linux/Unix
- ☐ Amazon Linux
- ☐ CentOS
- ☐ Debian
- ☐ Fedora
- ☐ Gentoo
- ☐ macOS
- ☐ openSUSE
- ☐ Other Linux
- ☐ Red Hat
- ☐ SUSE Linux
- ☐ Ubuntu

## ▼ Windows

- ☐ All Windows

## ▼ Architecture

- ☐ 64-bit (Arm)
- ☐ 32-bit (x86)
- ☐ 64-bit (x86)
- ☐ 64-bit (Mac)
- ☐ 64-bit (Mac-Arm)

## ▼ Root device type

- ☐ EBS
- ☐ Instance store

## ▼ ENA enabled

- ☐ ENA enabled

## ▼ Verified provider

- ☐ Verified provider

# AMI – Boot Modes

AMIs have two possible boot modes:

## Legacy BIOS (Basic Input/Output System)

The traditional firmware interface for computers, which has been used for decades.

It initializes hardware during the boot-up process and provides runtime services for operating systems and programs.

- No support for secure boot
- May be require when using legacy OS or legacy Software
- **Being replaced by UEFI in most modern workloads**

## Unified Extensible Firmware Interface (UEFI)

A modern firmware interface for computers, designed to replace the older BIOS firmware interface.

- Supports secure boot
- Faster startup times
- Supports drives larger 2TB
- Pre-boot environment with Graphical UI and some network capabilities



Unless there is a specific legacy reason you want to utilize UEFI

# AMI - Elastic Network Adapter (ENA)

Elastic Network Adapter (ENA) supports **network speeds of up to 100 Gbps** for compatible instance types.

Provides high-performance networking for EC2 instances



Amazon Linux 2023 AMI

ami-0440d3b780d96b29d (64-bit (x86), uefi-preferred) / ami-0f93c02efd1974b8b (64-bit (Arm), uefi)

Amazon Linux

Free tier eligible

Verified provider

Amazon Linux 2023 is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Platform: amazon      Root device type: ebs      Virtualization: hvm      ENA enabled: Yes

**Select**

☒ 64-bit (x86), uefi-preferred  
☐ 64-bit (Arm), uefi

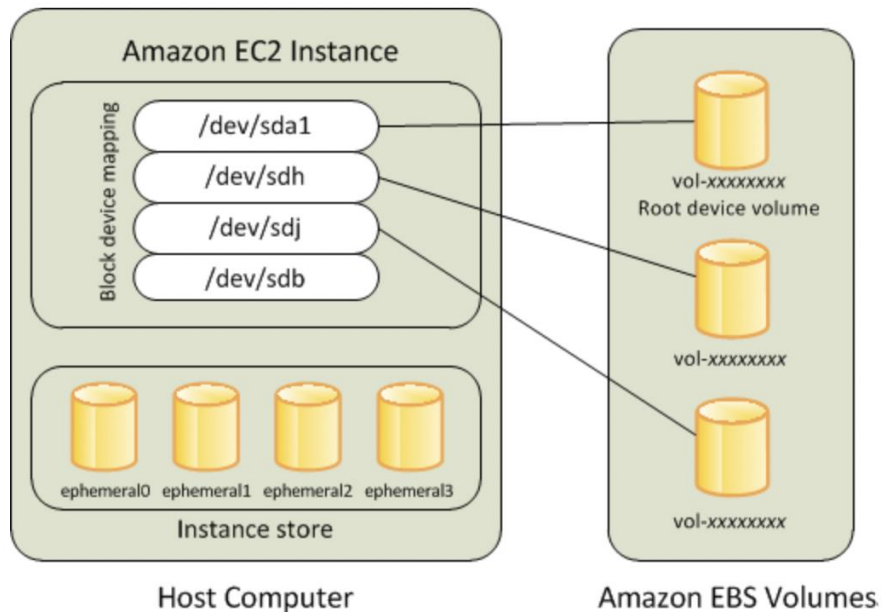
AMI's may have **ENA supported**

# Root Device Type

An AMI root device type can be either EBS-backed or Instance Store-backed

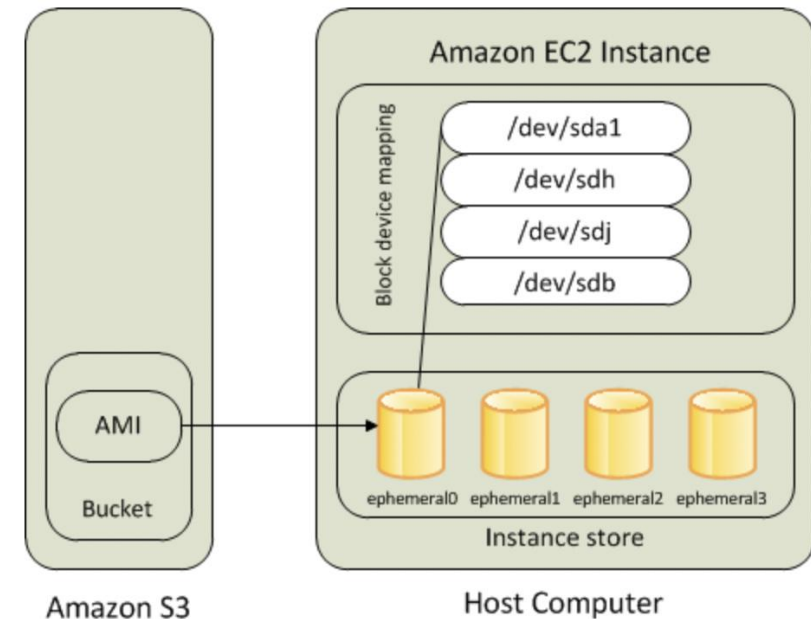
## Amazon EBS-backed instances

EBS volume is automatically attached at time of launch.  
The **storage is independent of the instance** and the instance be stopped or terminated with loss of data



## Instance store-backed instances

The native volumes of the instances are used.  
When the instance is stopped or terminated all data will be loss.  
Phased out for most use cases.



# AMI – Creating and Copying

```
aws ec2 create-image \  
--instance-id i-1234567890abcdef0 \  
--name "My server AMI" \  
--reboot
```

You can **create an AMI** from an existing EC2 instance that's either **running** or **stopped**.

```
aws ec2 copy-image \  
--source-region us-west-1 \  
--source-image-id ami-1234567890abcdef0 \  
--name "My copied AMI" \  
--encrypted \  
--region us-east-1
```

You can copy an AMI across regions for disaster recovery or global distribution  
You can enable encryption during copying (optional)

# AMI – Store and Restore

AMIs are stored in an Amazon S3 bucket (managed by AWS) and restore from the S3 Bucket.

You would do this if you want to copy AMIs from one AWS **partition** to another

Why would use use this over copy?

- Cheaper long-term storage
- Lifecycle management of AMIs
- Disaster recovery
- Auditing and compliance

## Storing an AMI to S3

```
aws ec2 create-instance-export-task \  
--instance-id i-1234567890abcdef0 \  
--target-environment vmware \  
--export-to-s3-task DiskImageFormat=VMDK,S3Bucket=my-ami-backup-bucket,S3Prefix=exported-ami/
```

## Restoring an AMI from S3

```
aws ec2 import-image \  
--description "Imported AMI" \  
--disk-containers "Format=VMDK,S3Bucket=my-ami-backup-bucket,S3Key=exported-ami/my-exported-vm.vmdk"
```

# AMI – Deprecate, Disable and Deregister

```
aws ec2 deregister-image --image-id ami-1234567890abcdef0
aws ec2 delete-snapshot --snapshot-id snap-1234567890abcdef0
```

- **Deregistering** is when you're done using an AMI and you don't want to allow any new instances to be launched with the AMI
- Deregistering **does not delete snapshots**

```
aws ec2 enable-image-deprecation \
--image-id ami-1234567890abcdef0 \
--deprecate-at "2022-10-15T13:17:12.000Z"
```

**Deprecating** an AMI allows you to set a **future expiration date** after which the AMI cannot be used

```
aws ec2 disable-image \
--image-id ami-1234567890abcdef0
```

**Disabling** temporarily blocks the use of the AMI. Cannot be re-enabled later

# AMI – Sharing

AMIs can have the following settings

- Public — Accessible by all AWS accounts
- Explicit — Shared with specific AWS Accounts, Organizations, or Organizational Units (Ous)
- Implicit — Available only to the AMI owner

```
aws ec2 modify-image-attribute \  
  --image-id ami-0abcdef1234567890 \  
  --launch-permission "Add=[{Group=all}]"
```



Shared AMIs can also be listed on AWS Marketplace

# AMI - Virtualization Types

|                   | Hardware Virtual Machine (HVM)                                                                              | Paravirtualization (PV)                                                                                                |
|-------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Virtualization    | Full virtualization with hardware assistance.                                                               | Software-assisted virtualization requiring OS modification.                                                            |
| Hardware Assisted | uses hardware-assist technology from the host system's CPU.                                                 | relies on a hypervisor to simulate hardware.                                                                           |
| Performance       | Potentially higher, especially for applications needing direct access to hardware.                          | Initially had better performance for some workloads due to lower overhead, but has been surpassed by HVM advancements. |
| OS Support        | Broader, can run unmodified operating systems. Any OS that can run on the physical hardware can run on HVM. | Limited to OSes modified for paravirtualization. Cannot run unmodified operating systems.                              |
| Boot Method       | Can boot from an EBS volume or an instance store.                                                           | Can only boot from an instance store.                                                                                  |
| Usage             | Recommended for modern operating systems and applications requiring specific hardware features.             | Historically used for certain workloads and older instance types. Less commonly used for new deployments.              |